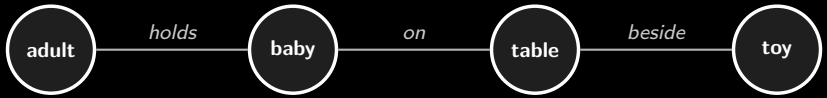
REVIEW 3 ▪ VIDOR DATASET

Dynamic Scene Graph Generation from Video

Detect objects, track them through time, and predict how they relate — frame by frame.



VidOR parts 1–8 trained ▪ 78-class pipeline

A video is a story of who does what to whom



Real validation frames with detector output (run 3)

OUTPUT PER FRAME
(illustrative)

adult *holds* **baby**

baby *looks at* **adult**

adult *next to* **baby**

Doing this for every frame, with objects that keep their identity over time, is the task this project tackles.

Per-frame graphs miss what changes over time

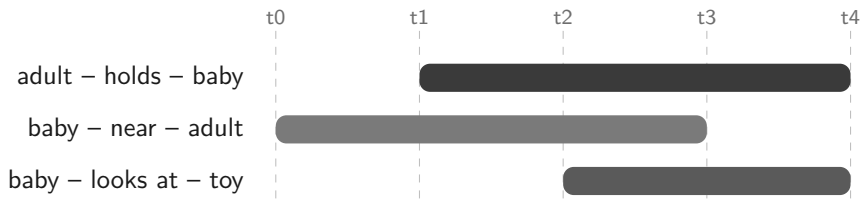
Static (per frame)

- Each frame labelled on its own
- No identity link between frames
- Relations flicker on and off

Dynamic (this project)

- Tracked objects keep one ID
- Relations have a start and an end
- Context from neighbouring frames

RELATIONS AS INTERVALS



VidOR comes in eight parts — all eight are now trained



Dark = trained earlier Grey = added for this review

1 fps frame sampling

60,588 training frames

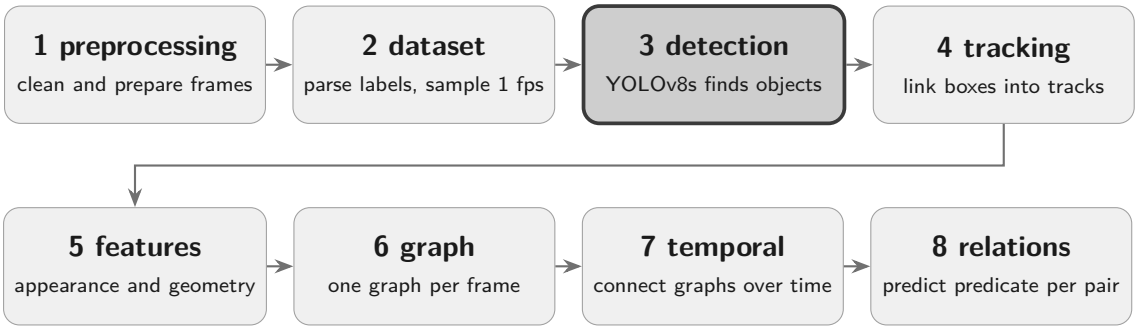
3,594 validation frames

+750 new videos (3 groups)

Training frames = 33,277 earlier + 27,311 new.

Same validation set as before.

Eight stages from raw video to scene graph



Trained and evaluated: detection, the relation model and the temporal LSTM/GRU models. Later stages consume the detector's boxes, so detector quality sets the ceiling.

What the dataset export contains

Folder / file	Contents
`annotations`	VidOR labels for train and val
`videos`	source clips
`yolo_format`	frames and boxes in YOLO layout
`tracks`	per-video object tracks
`features`	extracted object features
`relation_dataset`	object-pair relation samples
`relation_dataset_corrected`	revised relation samples
`dataset_manifest.json`	parts, fps and counts

YOLOv8s: three Colab runs, then a fine-tune on all parts

Base model	**yolov8s.pt**, pretrained
Input size	**640 px**
Epochs	**10** per run, **3** fine-tune
Optimizer	auto, lr0 = 0.01
Augmentation	mosaic, flip 0.5, HSV
Hardware	Colab GPU, then M3 Pro Mac

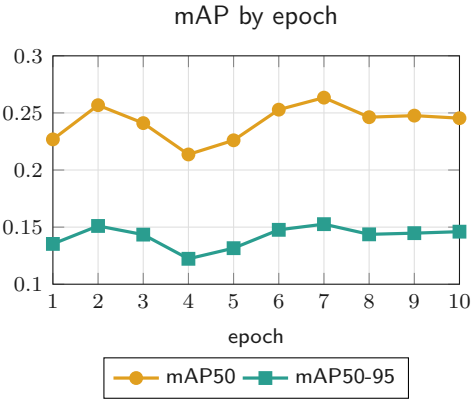
Run 1 batch 16 weights saved

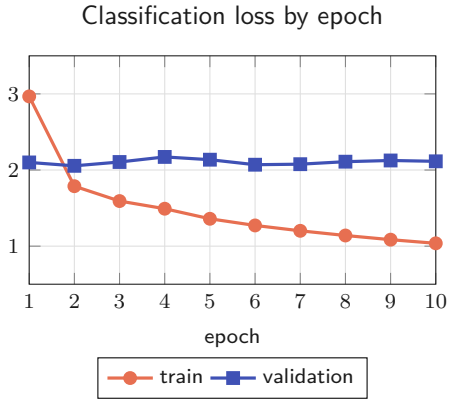
Run 2 batch 32 weights saved

Run 3 batch 32 best mAP50 0.263

Fine-tune batch 16 from run 3, all parts

mAP stalls near 0.25 as validation loss stays flat





Training loss keeps falling while validation loss stays near 2.1 — a sign of over-fitting. Full run: 10 epochs, about 79 minutes on one Colab GPU.

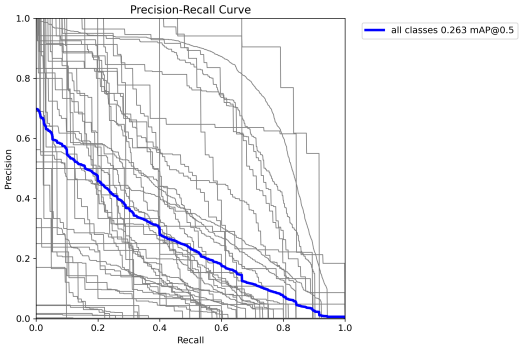
Peak 0.263 mAP50 at epoch 7

0.263
mAP50, best (ep. 7)

0.153
mAP50-95, best

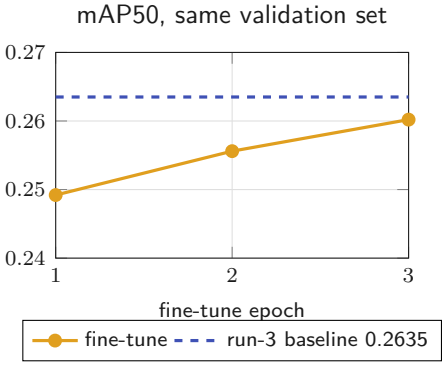
0.363
precision, final

0.251
recall, final



Precision–recall: all classes (blue) and each class (grey).

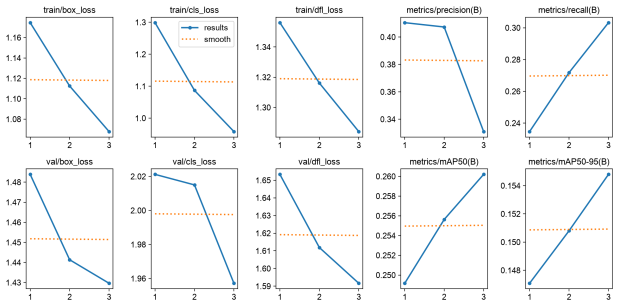
Fine-tune on all parts: detector results



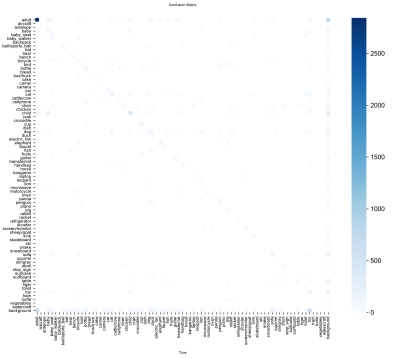
	P	R	mAP50	mAP50-95
Baseline (run 3)	0.381	0.285	0.2635	0.1526
Epoch 1	0.410	0.235	0.2492	0.1471
Epoch 2	0.407	0.272	0.2556	0.1508
Epoch 3	**0.331**	**0.303**	**0.2602**	**0.1548**

27,311 new frames did not raise mAP50 (0.2635 → 0.2602); recall rose and precision fell, mAP50-95 is slightly higher. Validation classification loss improved from about 2.1 to 1.96. Each epoch took about 76 minutes on an M3 Pro.

Fine-tune training graphs

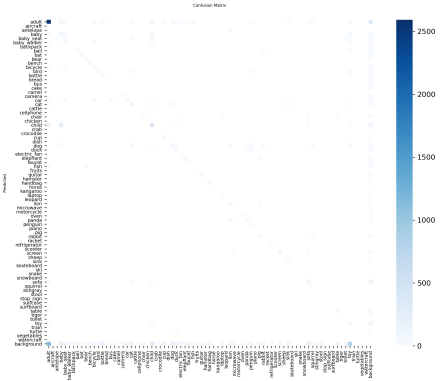


Losses, precision, recall and mAP over the 3 fine-tune epochs.



Confusion matrix after fine-tuning.

Strong on adults, thin on most other classes

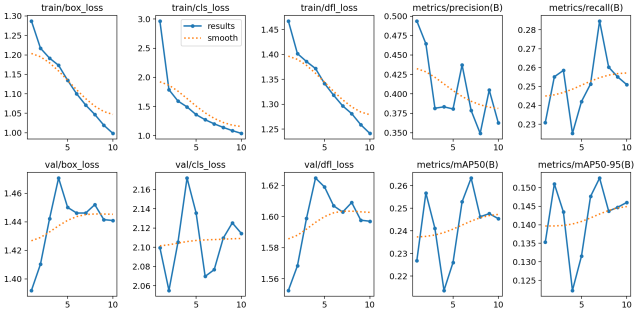


Confusion matrix: a dark adult cell, faint scatter elsewhere.

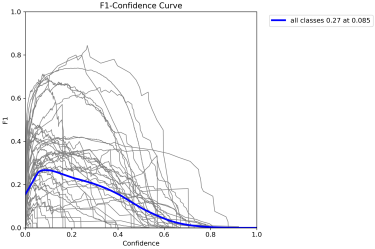


Validation predictions: adult and baby found, confidence 0.3–0.5.

Training graphs from the run

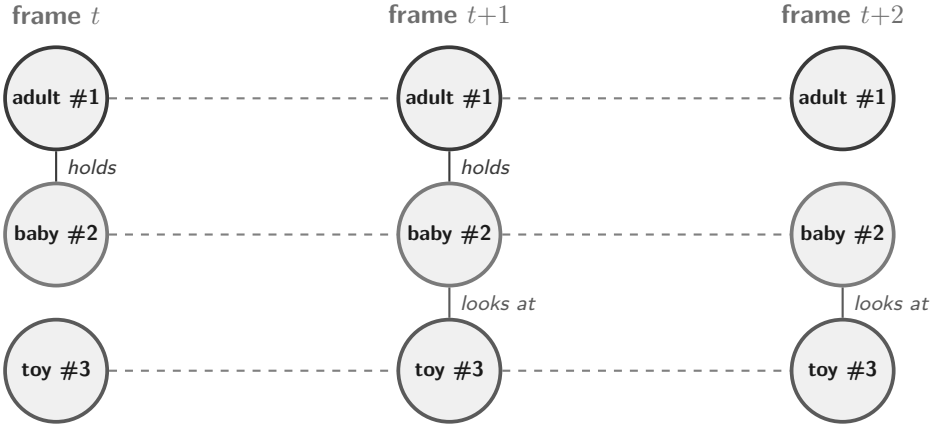


Losses, precision, recall and mAP over the 10 epochs.



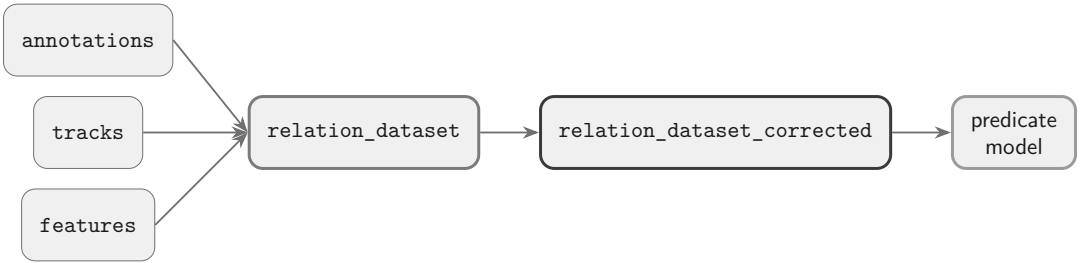
F1 versus confidence threshold.

Tracks turn per-frame boxes into a temporal graph



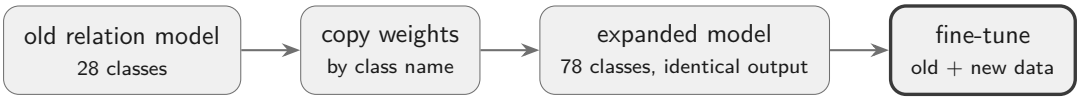
Solid edges: relations inside a frame. Dashed edges: the same object linked through time.

Relation samples come from tracks and features



Two exports exist: the first relation set and a corrected version. The corrected one is the cleaner input for relation learning.

Combining old and new training: 28 to 78 classes



	Old data	New data	Combined
Relation training pairs	7,536	145,031	152,567
Temporal training sequences	1,890	38,388	40,278

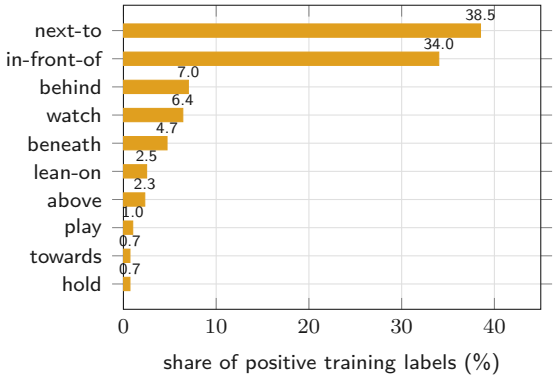
The expanded model gave exactly the same answers as the old one on the old validation set (accuracy 0.429, loss 1.758), so no knowledge was lost. 38 new videos (every 10th) were held out of training to test on unseen data.

Relation and temporal models: before → after fine-tuning

Model	Test set	Accuracy	Macro-F1	Loss
Relation	old val	0.429 → 0.529	0.068 → 0.077	1.758 → 1.490
Relation	new videos	0.303 → 0.407	0.098 → 0.124	2.264 → 1.781
Temporal LSTM	old val	0.645 → 0.586	0.072 → 0.099	1.317 → 1.375
Temporal LSTM	new videos	0.278 → 0.372	0.026 → 0.074	2.100 → 1.891
Temporal GRU	old val	0.576 → 0.628	0.075 → 0.113	1.510 → 1.372
Temporal GRU	new videos	0.281 → 0.390	0.032 → 0.063	2.284 → 1.809

- Every model improves on the unseen new videos; the old models had never seen most of the 78 classes.
- The LSTM is slightly worse on the old validation set (accuracy 0.645 → 0.586); the GRU improves on both.
- Scores are still low: macro-F1 is at most 0.13, because background pairs dominate the data.
- Validation loss is best after 1 to 3 epochs, then rises (over-fitting), so early stopping chose the checkpoints.

Why two predicates dominated the graphs



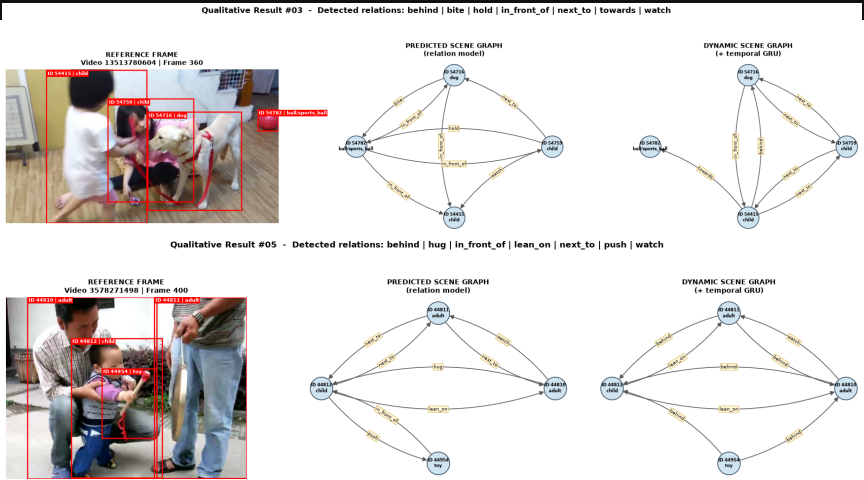
- **Data:** next_to and in_front_of are **72%** of all positive labels; the rarest predicates have fewer than 30 samples.
- **Loss:** v1 used an unweighted loss, so guessing the common classes pays off.
- **Checkpoints:** v1 was chosen by validation loss, but macro-F1 on new videos was still rising (0.124 $\rightarrow$ 0.156 by epoch 5).
- **Result:** v1 predicted only 11 of 32 predicates (the GRU only 4).

The fix: class-balanced loss, checkpoints chosen by macro-F1

Model	Test set	Macro-F1	Accuracy	Predicates used	Top-2 share
Relation	new videos	0.124 → 0.189	0.407 → 0.406	11 → 23	74% → 58%
Relation	old val	0.084 → 0.064	0.529 → 0.314	8 → 14	73% → 53%
Temporal LSTM	new videos	0.074 → 0.145	0.372 → 0.339	8 → 17	95% → 54%
Temporal LSTM	old val	0.099 → 0.059	0.586 → 0.322	4 → 10	92% → 47%
Temporal GRU	new videos	0.063 → 0.172	0.390 → 0.333	4 → 22	99% → 45%
Temporal GRU	old val	0.113 → 0.130	0.628 → 0.396	3 → 13	100% → 49%

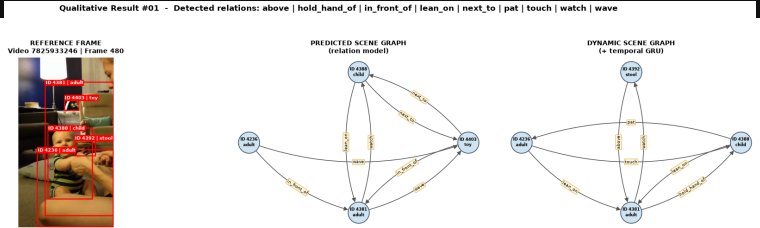
- v2 = v1 retrained with loss weights $\propto 1/\sqrt{\text{frequency}}$ (capped at 12×), checkpoint chosen by macro-F1. "Top-2 share" = share of non-background predictions that are next_to or in_front_of.
- On new videos every model gains macro-F1 and uses far more predicates; the GRU's top-2 share falls from 99% to 45%.
- Cost: accuracy drops (it rewarded guessing common classes). On the small old validation set the result is mixed: relation worse, LSTM worse, GRU better.

Dynamic scene graphs after the fix (1/2)



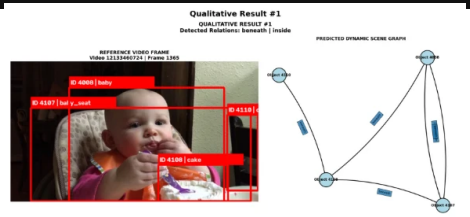
Held-out new videos. Examples were picked for predicate variety, not at random.

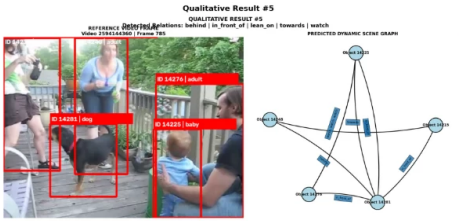
Dynamic scene graphs after the fix (2/2)

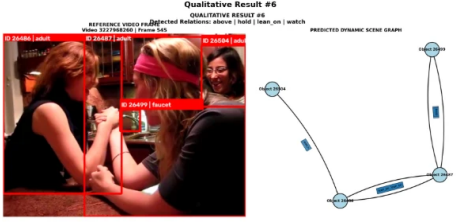


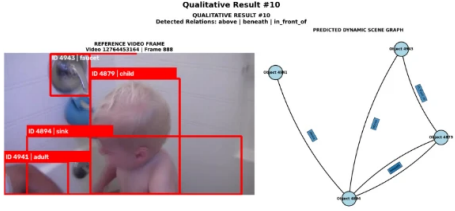
Some relations are plausible (watch, hold, towards), others are wrong: scores remain low.

Qualitative results: frames and predicted graphs









Left: tracked, labelled objects. Right: predicted scene graph with relation labels.

What slowed us down and what we learned

1 Compute

Limited GPU access: the final fine-tune ran on a Mac, about 4 hours for 3 epochs.

2 Class mismatch

The pipeline had 28 classes, the detector 78. Fixed by expanding by class name.

3 Over-fitting

Relation and temporal validation loss rises after 1 to 3 epochs.

4 Low scores

Detector recall is 0.30 and macro-F1 is at most 0.13.

Review 3: VidOR training progress



Stage	Before (parts 1–5)	After (parts 6–8 added)
Detector mAP50 / mAP50-95	0.2635 / 0.1526	0.2602 / 0.1548
Detector recall	0.285	0.303
Relation, new videos (acc / macro-F1)	0.303 / 0.098	0.408 / 0.124
Temporal GRU, new videos (acc / macro-F1)	0.281 / 0.032	0.390 / 0.063

Honest summary: more data did not raise detector mAP50; the relation and temporal models improved on unseen new videos, but scores remain low.

Next: evaluate end to end, then improve

1

Run the full pipeline

Evaluate detection to relations end to end

2

Train the detector longer

More epochs and a tuned learning rate

3

Weight the classes

Class-weighted relation loss to lift macro-F1

4

Use every video

Track and featurise all new videos, not every second one

Thank you